

Tommaso Virgili

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EDUCATION

Sapienza University of Rome <i>Master of Science in Condensed Matter Physics</i>	Rome, Italy Sep. 2025 – Present
Sapienza School for Advanced Studies (SSAS) <i>Honors College of Sapienza University of Rome</i>	Rome, Italy Oct. 2025 – Present
Sapienza University of Rome <i>Bachelor of Science in Physics, Grade: 110/110 cum laude</i>	Rome, Italy Sep. 2022 – Sep. 2025
Liceo Classico Augusto <i>High School Diploma, Classical Studies, Grade: 100/100 cum laude</i>	Rome, Italy Sep. 2017 – Jul. 2022

EXPERIENCE & RESEARCH

Lab Internship for examination Nanospectroscopy Lab <i>Sapienza University of Rome (Prof. M. De Luca)</i>	Mar. 2026 – Jul. 2026 Rome, Italy
- Gained hands-on experience in sample preparation and operation of optical spectroscopy setups for semiconductor nanostructures characterization (nanowires). - Operated high-precision sub-micron piezoelectric micro-positioners. - Performed data analysis on the spectra with physical interpretation. - Collaborated effectively within a lab team, managing task allocation and workflow. - Achieved maximum grade for the activity: 30/30.	
B.Sc. Thesis – White Rabbit Synchronization System <i>Sapienza University of Rome (Advisor: Prof. C.A. Nicolau)</i>	Jun. 2025 – Sep. 2025 Rome, Italy
- Investigated sub-nanosecond precision timing distribution networks for large-scale physics experiments (scale of 100 m to 1000 m), with a focus on the KM3NeT underwater neutrino telescope. - Authored review thesis bridging electronic engineering and high-energy experimental physics.	

TECHNICAL SKILLS

Programming & Software <i>C, Python, Bash, GNU/Linux, L^AT_EX</i>	
- Proficient in scientific programming in C and scientific data analysis using Python (NumPy, SciPy, Matplotlib, Pandas). - Strong Linux environment familiarity, shell scripting, and system-level hardware troubleshooting. - Experience in software interfacing with experimental laboratory instrumentation. - Designed and maintain a personal NON technical website/blog (tomtomvir.github.io).	
Electronics & Laboratory Skills <i>Instrumentation, Circuit Troubleshooting, Soldering</i>	
- Hands-on experience with electronic circuit debugging, soldering, and prototyping. - Familiar with standard electronic measurement instrumentation (oscilloscopes, multimeters, function generators).	
Optics & Spectroscopy Skills <i>Free-Space Optics, Lasers, Optical Alignment, Piezo Positioning</i>	
- Hands-on experience with free-space optical setups, beam alignment, lens/mirror systems, beam splitters, and spectral filters. - Operation of spectroscopy systems (e.g., Raman, Photoluminescence) for characterization of semiconductor nanostructures. - Sub-micron spatial positioning using piezoelectric actuators for precise optical excitation and mapping. - Familiarity with optical instrumentation (laser sources, optical power meters, CCD detectors, spectrometers) and optical table safety.	

LANGUAGES & COMPLEMENTARY SKILLS

Languages: Italian (Native), English (Advanced / C1), Latin, Ancient Greek
Developer Tools: Git, GitHub, Linux CLI
Honors & Interdisciplinary Training: Student Fellow at Sapienza School for Advanced Studies (SSAS) — Highly selective merit-based honors program. Engaged in interdisciplinary coursework (e.g., Experimental Economics, Sociology) with emphasis on public speaking, team leadership, and delivering conference-style presentations.